

VAISHNAV RAJA

857-396-3939 | raja.krv@northeastern.edu | [linkedin.com/in/vaishnavraja/](https://www.linkedin.com/in/vaishnavraja/) | vaishnavraja11.github.io/portfolio/ | Boston, USA

EXPERIENCE

Philips

R&D Engineering Intern, Surgical Robotics

Cambridge, MA

May 2024 – Dec 2024

- Built a Python pipeline for **multi-arm workspace** feasibility: reachability, manipulability (Yoshikawa), singularity margins, collision buffers; plots/metrics for design decisions.
- Implemented analytical and **Jacobian DLS IK** with joint-limit handling and singularity proximity scoring; scripts for repeatable evaluations.
- Refactored $\sim 1600 \rightarrow 700$ LOC and accelerated sweeps with **multiprocessing & Numba**; packaged runs for CI-style reproducibility.
- Prototype **EtherCAT** tool-controller bring-up with central node; created lab test steps and sanity checks for reliable communication.

Tata Consultancy Services

Component Design Engineer, Infotainment Systems & Accessories

Chennai, India

May 2021 – Aug 2023

- Owned design/release for a **shark-fin antenna**; commonized across 4 regions; drawings/BOMs via ENOVIA; supplier coordination through SOP.
- Executed CAD/PCB checks (CATIA) and CAE (Simulia); documented design changes and validation results.
- Eliminated speaker noise & alternator whine via targeted **EMI/EMC** filters; stabilized audio path and cleared program quality gates.

PROJECTS

6D Pose Estimation for Industrial Bin-Picking

- Multi-view RGB-D pipeline: FoundationPose initialization with **ICP** refinement for known CAD parts under clutter/occlusion.
- Generated pose overlays & error summaries; batch scripts for repeatable runs and throughput profiling.

Goal-Oriented Robot Navigation with Deep RL

- Trained a differential-drive agent with **DDPG/TD3**; reward shaping for progress, heading, and collisions; logging hooks and batch evaluators for **quantitative benchmarking**.

UR5 — Kinematic Control for a Square Trajectory

- DH-based FK and iterative **Jacobian IK** (body frame) to track a 3D square with fixed tool orientation; verified with 3D animation.

Warehouse Automation: AGV + Pick-to-Light + Stock Management

- Integrated **Pick-to-Light** with an AGV and inventory system; Pi site controller; dual Arduino per-wheel PID and a custom stepper driver; reliable Pi $\leftrightarrow$ Arduino telemetry.
- Delivered real-time task dispatch, confirmations, and live status UI; **MongoDB**-backed persistence; customer demo recognized with Accenture **Best Innovator Award** (2019).

TECHNICAL SKILLS

Programming: C/C++, Python, Bash; Git/GitLab CI; Linux CLI

Robotics/Controls: ROS2, Gazebo; Kinematics; MATLAB/Simulink; EtherCAT; Mitsubishi MELFA, KUKA KRC

Perception/ML: OpenCV; RGB-D/Point Clouds; ICP refinement; 6D Pose Estimation; (RL: DDPG/TD3)

EE/Embedded: KiCad, SPICE; wiring/bring-up; EMI/EMC mitigation; custom stepper driver; Raspberry Pi, Arduino

Mech/CAD: SolidWorks, CATIA; basic machining & 3D printing; Ansys; DFM/DFX

EDUCATION

Northeastern University

Master of Science in Robotics

Boston, MA

Aug 2025

- Relevant: Mobile Robotics, Control Systems, Reinforcement Learning, Computer Vision, Algorithms, Robot Sensing & Navigation

Anna University

Bachelor of Engineering in Mechatronics Engineering

Chennai, India

Apr 2021

- Relevant: Industrial Automation, Sensors & Signal Processing, Kinematics & Dynamics of Machines, Mechatronics Systems Design